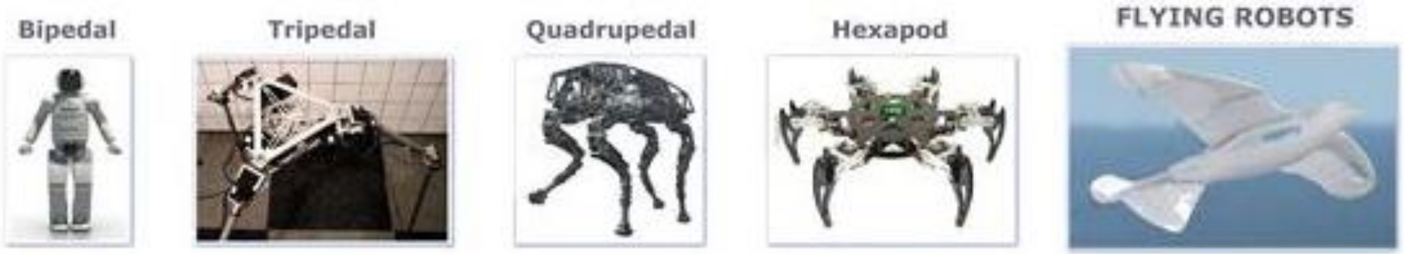

PATH PLANNING

PATH PLANNING

- Path planning of a mobile robot is one of the basic operations needed to implement the navigation of the robot.
- The path planning operation provides the answer to the question “how should I get to where I’ am going?”
- Path planning gives a feasible collision free path for going from one place to another.

- On the basis of the way the information about the robot's environment is obtained, most of the path planning methods can be classified into two categories:
  - Model-based approach

    In the model based approach, all the information about the robot's workspace are prelearned, and the user specifies the geometric models of objects and a description of them in terms of these models.
  - Model-free approach

    In the model-free approach, some of the information about the robot's environment is obtained via sensors (e.g., vision, range, touch sensors). The user has to specify all the robotic motions needed to accomplish a task.

MODEL-BASED ROBOT PATH PLANNING

- The obstacles that may exist in a robotic work environment are distinguished into static obstacles and moving obstacles. Therefore, two types of path finding problems have to be solved, namely:
- Path planning among stationary obstacles
- Path planning among moving obstacles

- The path planning methodology for stationary obstacles is based on the configuration space concept, and is implemented by the so-called road map planning methods.
- The path planning problem for the case of moving obstacles is decomposed into two subproblems:
  - Plan a path to avoid collision with static obstacles.
  - Plan the velocity along the path to avoid collision with moving obstacles.

CONFIGURATION SPACE

- Configuration space is a representation in which path planning for mobile robots is performed.
  - For example, if the mobile system is a free-flying rigid body (i.e., a body that can move freely in space in any direction without any kinematics constraint), then six configuration parameters are required to determine fully its position, namely, x; y; z and three directional (Euler) angles. Path planning finds a path in this six-dimensional space.
  - A robot is not a free-flying robot. Its possible motion depends on its kinematic structure,
  - for example, a unicycle-like robot has three configuration parameters: x; y, and φ.

- Given a robot with ‘n’ configuration parameters moving in a certain environment, we define the following:
  - The configuration q of the robot, that is, an n-tuple of real numbers that specifies the n parameters needed to determine the position of the robot in physical space.
  - The configuration space (CS) of the robot, that is, the set of values that its configuration q may take.
  - The free configuration space CS_{free}, that is, the subset of CS of configurations that are not in collision with the obstacles existing in the robot’s environment.
  - Path planning is the problem of finding a path in the free configuration space CSfree, between an initial and a final configuration

- Thus, if CSfree could be determined explicitly, then path planning is reduced to a search for a path in this n-dimensional continuous space.
- Actually, the explicit definition of CSfree is a computationally difficult problem (its computational complexity increases exponentially with the dimension of CS), but there are available efficient probabilistic methods that solve this path planning problem in reasonable time.
- These techniques must involve two operations:
  - Collision checking (i.e., check whether a configuration q or a path between two configurations lies entirely in CSfree)
  - Kinematic steering (i.e., find a path between two configurations q0 and qf in CS that satisfies the kinematic constraints, without taking into account obstacles)